Sanaka Satyanarayana

KAVACH V&V Designer | STCAS - Application Logic Development for FTP/FAT/SAT | Functional & Developmental Testing

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PROFILE

- Railway Signalling Design Engineer with 3+ years of hands-on experience in TCAS system design and STCAS application logic configuration. Proven expertise in end-to-end signalling data preparation, including TOC, Tag Data, and Track Profiles, along with execution of FTP, FAT, and SAT.
- Skilled in interface configuration involving S-KAVACH to S-KAVACH communication (RRI data exchange) and PCCL documentation. Collaborated with RDSO and CoE/IRISET teams to ensure compliance with Indian Railways standards.
- Focused on delivering reliable and compliant ATP solutions across complete project lifecycles—from design to commissioning.

PROFESSIONAL EXPERIENCE

01/2025 – present
Hyderabad

KAVACH V&V Designer
Quadrant FutureTek

- Developed and configured STCAS application logic for Kavach system validation, covering HoC, ReC, SPAD prevention, turnout speed control, and SOS scenarios under diverse field conditions.
- Designed VDU screens for enhanced operational visibility and system monitoring.
- Led end-to-end configuration, integration, and deployment of S-Kavach across 7 stations (SCR), from initial setup through field validation, including SAT, passenger trials, and interoperability testing at IRISET.
- Designed 250+ FTP test cases and conducted system-level V&V activities, ensuring requirement traceability, validation coverage, and system compliance.
- Performed STCAS developmental testing and finalized initial configuration parameters using dynamic profiles.
- Collaborated with cross-functional teams (HW/SW/Site) to resolve issues, validate fixes, and maintain alignment with safety and system requirements.
- Configured inter-station data exchange and validated build artifacts, ensuring seamless train handover and robust hardware–software integration.

08/2023 – 12/2024
IRISET, Secunderabad

Technical Expert - Research & Development in KAVACH
Center of Excellence (Modern Signalling, Kavach) / IRISET - Indian Railways

- Simulation, modeling, monitoring and managing the performance and availability and creation of Railway systems under development.
- Performed performance analysis of Kavach systems to evaluate reliability and efficiency.
- Conducted root cause analysis of Kavach incidents to identify system-level issues and improvements.
- Involved in evaluation process of Expression of Interest (EoI) issued by Centre of Excellence (CoE)/IRISET - Indian Railways.
- Collaborating with subject matter experts and engineers to gather information and validate technical accuracy and adoption of global best practices.
- Involved in testing Railway specific 5G Use Cases in collabration with IIT-M.

03/2023 – 08/2023
Rail Nilayam,
Secunderabad

KAVACH (TCAS) System – Verification & Validation Exposure
Rail Nilayam, South Central Railway - Indian Railways. (Self-driven learning)

- Gained hands-on exposure to Indian Railways signalling systems and operations
- Performed signalling data verification including TOC, Track Profiles, and RFID (TIN) layouts against field data
- Supported design validation against field signalling layouts
- Acquired knowledge of FAT/SAT processes, simulator-based testing, and RDSO/IRISET standards.

PROJECTS

03/2023 – present

Train Collision Avoidance System (TCAS)
KAVACH - IR (ATP)
The Kavach system offers monitoring and control of train movement, constant supervision of speed, and automatic braking in case of any potential danger. It comes with train and track-side sensors, communication systems, track-side and on-board vital processors compliant to SIL 4 architecture, and algorithms that quickly detect and respond to any threat to train safety. The technology behind Automatic Train Protection Systems can be classified into two categories based upon mechanism of exchange of information between trackside and train units -Radio based and non-radio based. Kavach-Cab signalling & Automatic Train Protection system is a Radio based system designed to enhance train traffic throughput and adaptable to newer functionality to meet the future roadmap of Indian Railways.

CERTIFICATES

Kavach Training Certificate

Specialized on-the-job training on "Automatic Train Protection - Kavach (KAVACH/0018)" held from 11th Sept 2023 to 06th Oct 2023 i.e. 4 Weeks in the campus of Indian Railways Institute of Signal Engineering and Telecommunications - IRISET : S&T Centralized Training Institute, Government of India – Ministry of Railways.

AutoCAD
Certified in AutoCad from National Skill Development Institute, Hyderabad.

EDUCATION

2018 – 2022
Seshadri Rao Gudlavalleru Engineering College
B. Tech

2016 – 2018
Narayana Junior College
Intermediate

SKILLS

- Modern Railway Signalling - Kavach
- Kavach Verification and validation (V&V)
- VDU Design.
- MS Office
- STCAS Application Logic Development
- European Train Control System (ETCS)
- Communications Based Train Control (CBTC)
- AutoCAD

LANGUAGES

- English
- Telugu
- Hindi

DECLARATION

I do here by declare that the particulars of Information stated above is true, correct and complete to the best of my knowledge and belief.

Sanaka Satyanarayana
Hyderabad